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Lin

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(54) **SAFETY STRAP**

(71) Applicant: **Chung-Kun Lin**, Chiayi (TW)

(72) Inventor: **Chung-Kun Lin**, Chiayi (TW)

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(51) **Int. Cl.**

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D03D 1/00	(2006.01)
D03D 15/63	(2021.01)
D03D 13/00	(2006.01)

(52) **U.S. Cl.**
CPC ***D03D 15/63*** (2021.01); ***D03D 1/0005*** (2013.01); *D10B 2403/023* (2013.01)

(58) **Field of Classification Search**
CPC . D03D 15/63; D03D 1/0005; D10B 2403/023
See application file for complete search history.

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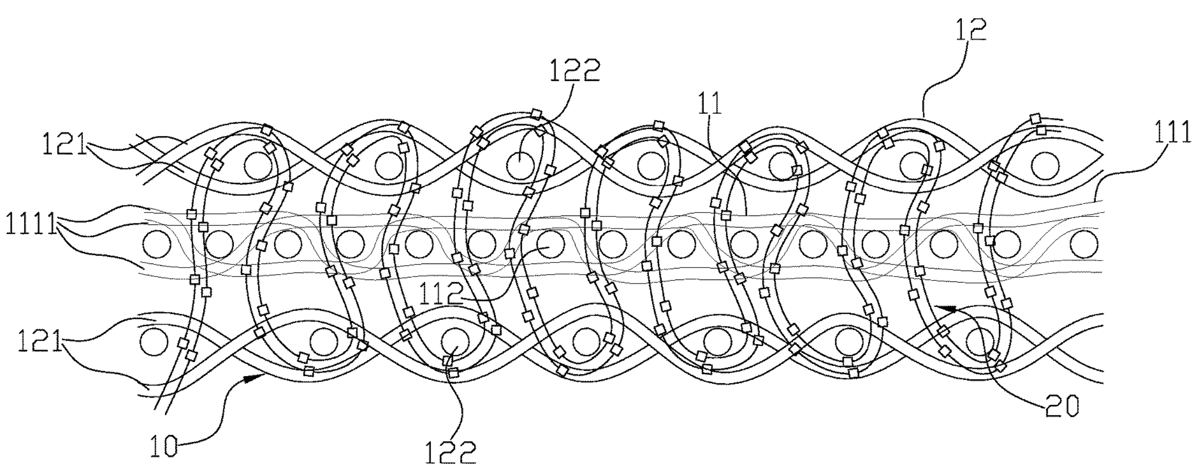
Primary Examiner — Robert H Muromoto, Jr.

(57) **ABSTRACT**

A safety strap has an outer layer, an inner layer disposed in the outer layer and a plurality of interlace strands. The inner layer has a plurality of warp strand bundles and a plurality of first weft strands woven with each other. Each of the warp strand bundles has more than one first warp strand. The outer layer is woven with a plurality of second warp strands and a plurality of second weft strands. The interlace strands continuously intertwine with the second weft strands and the first weft strands.

3 Claims, 9 Drawing Sheets



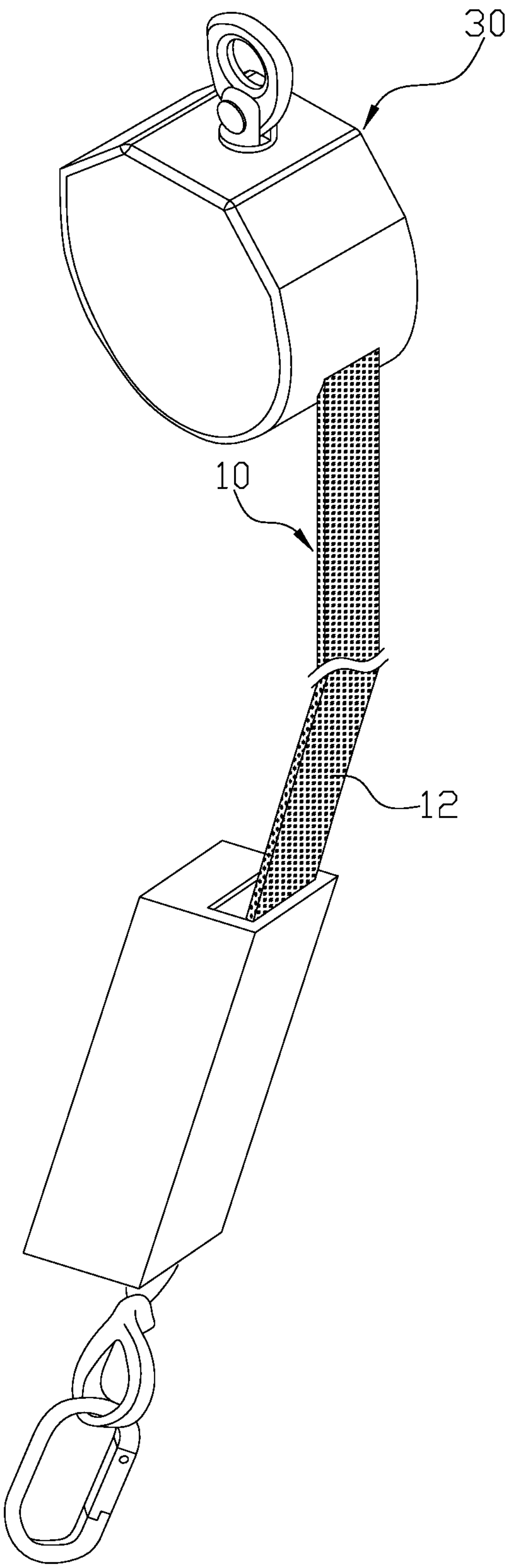


FIG. 1

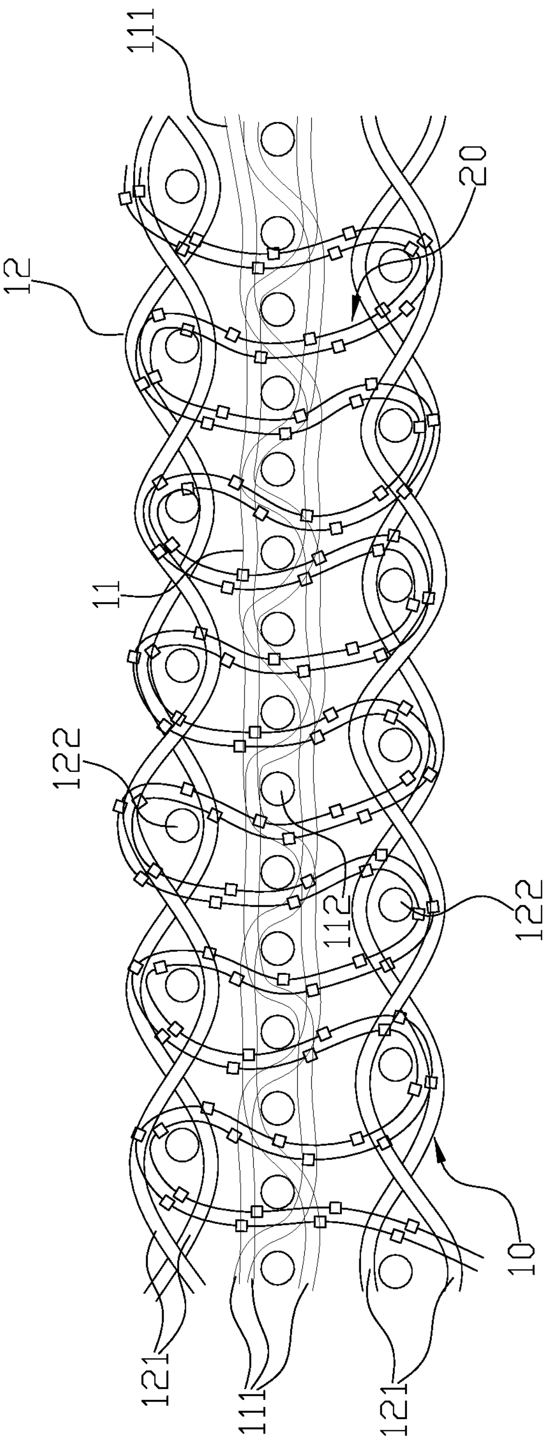


FIG. 2

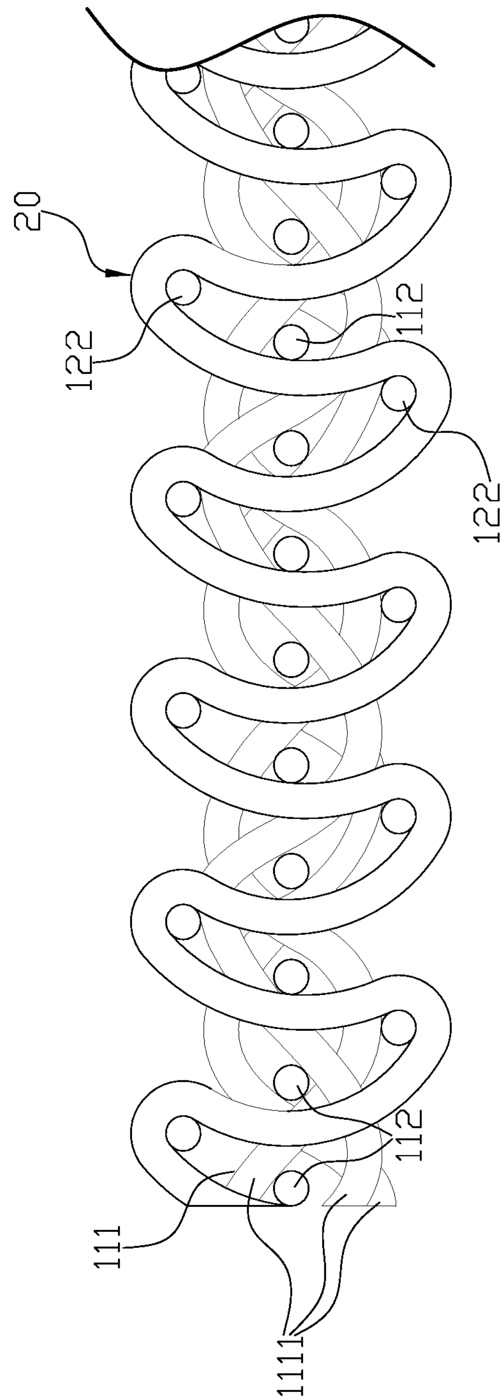


FIG. 3

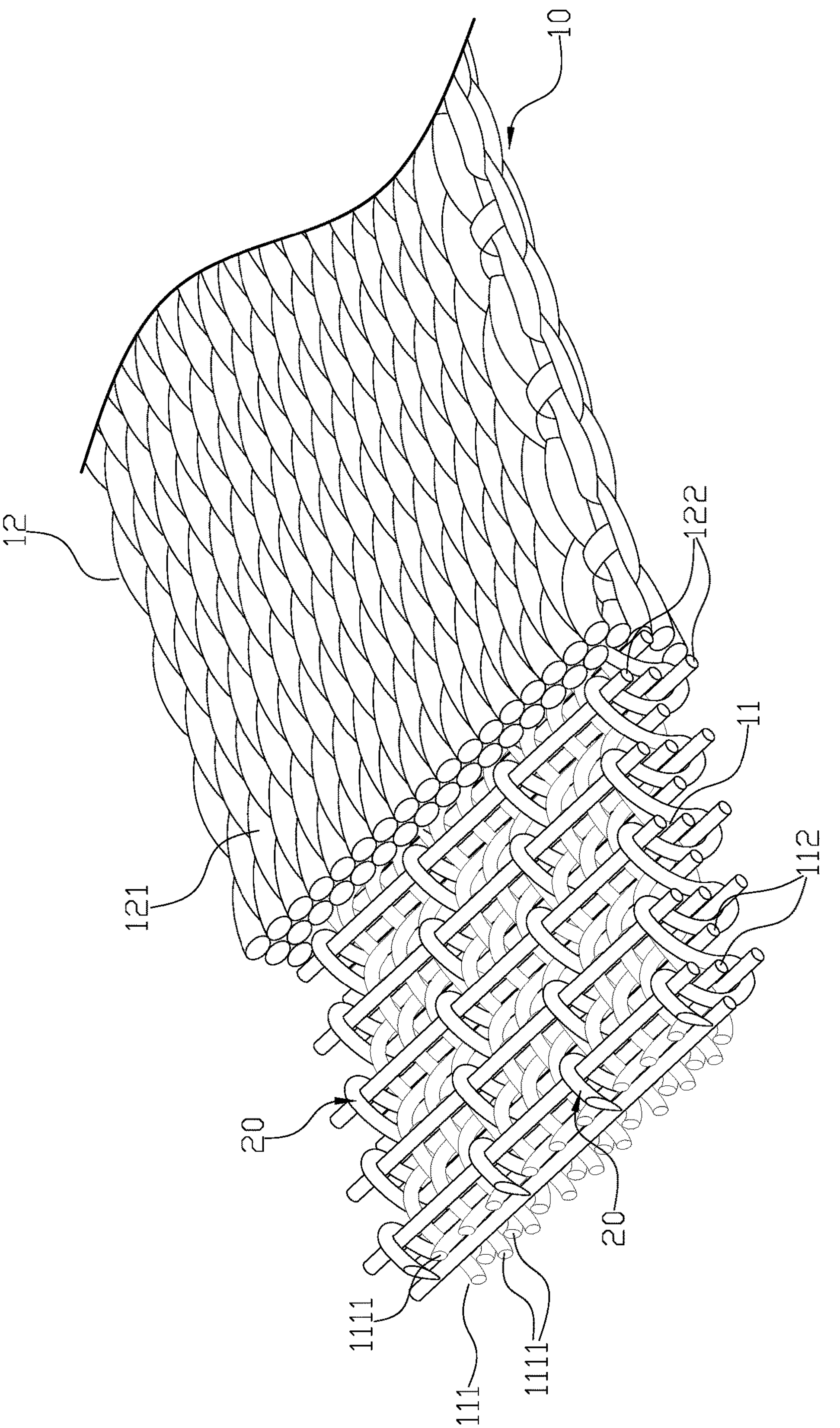


FIG. 4

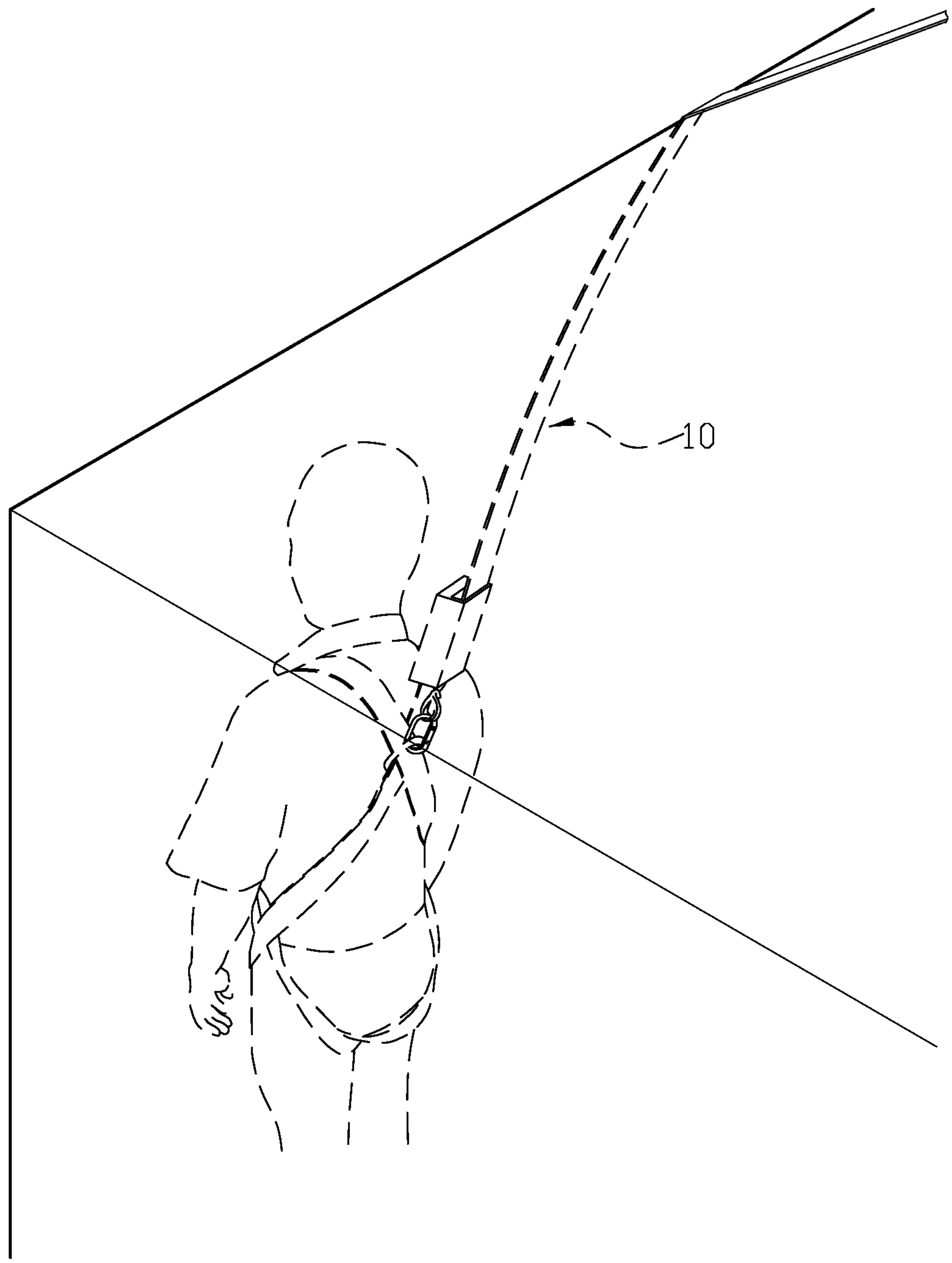


FIG. 5

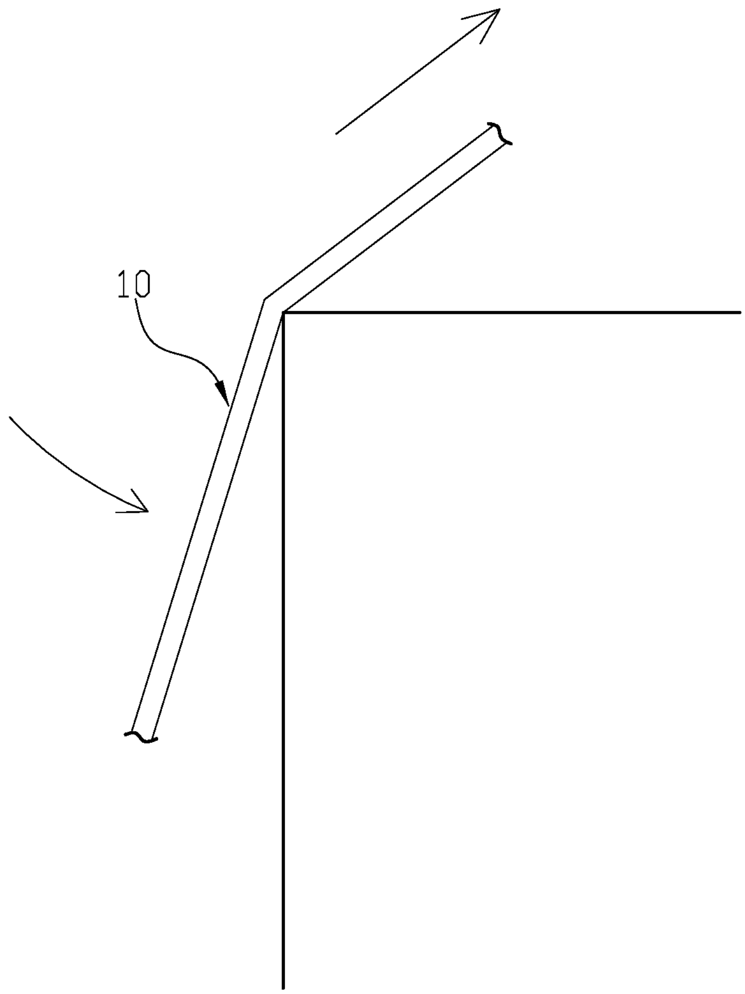


FIG. 6

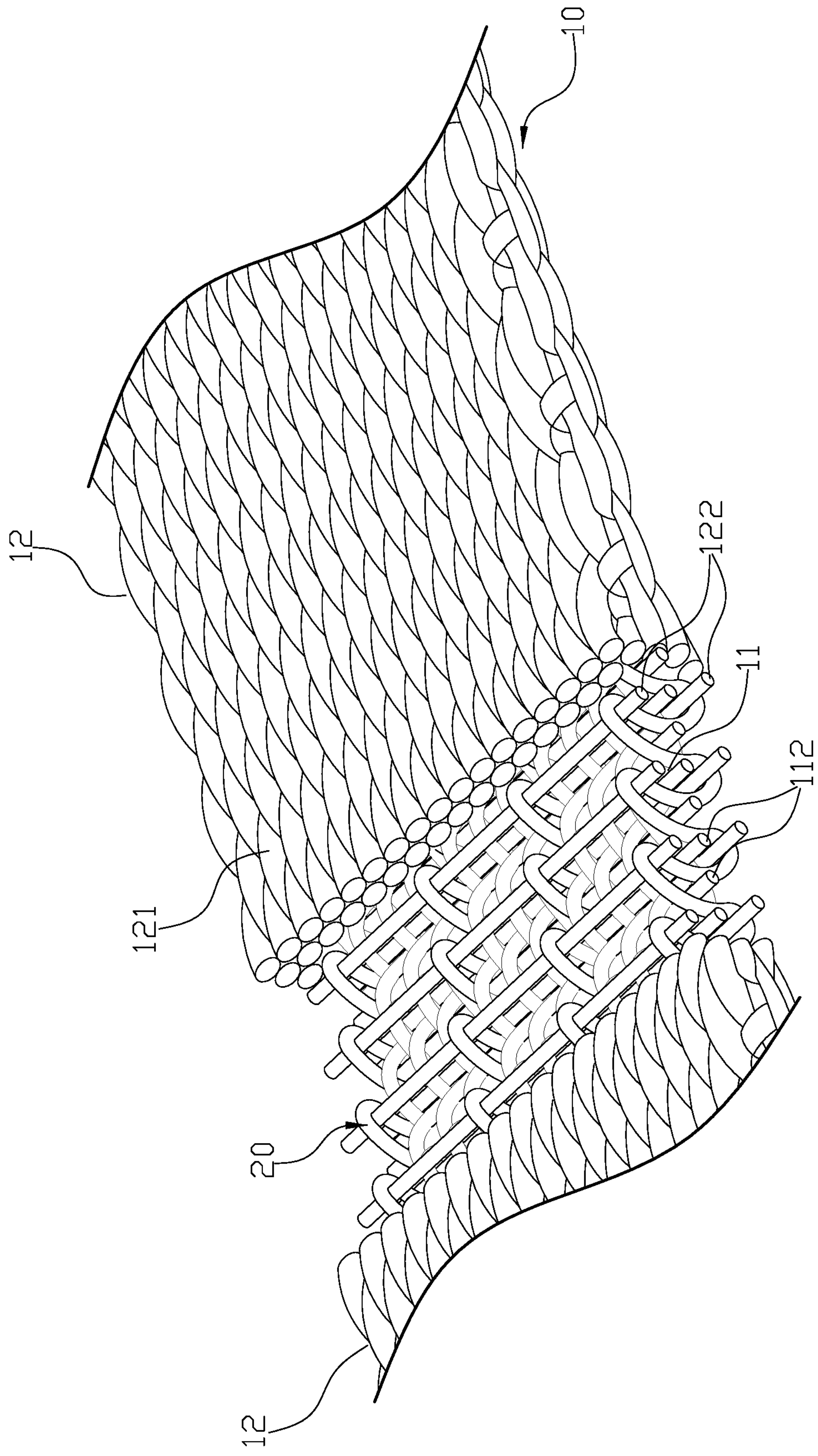


FIG. 7

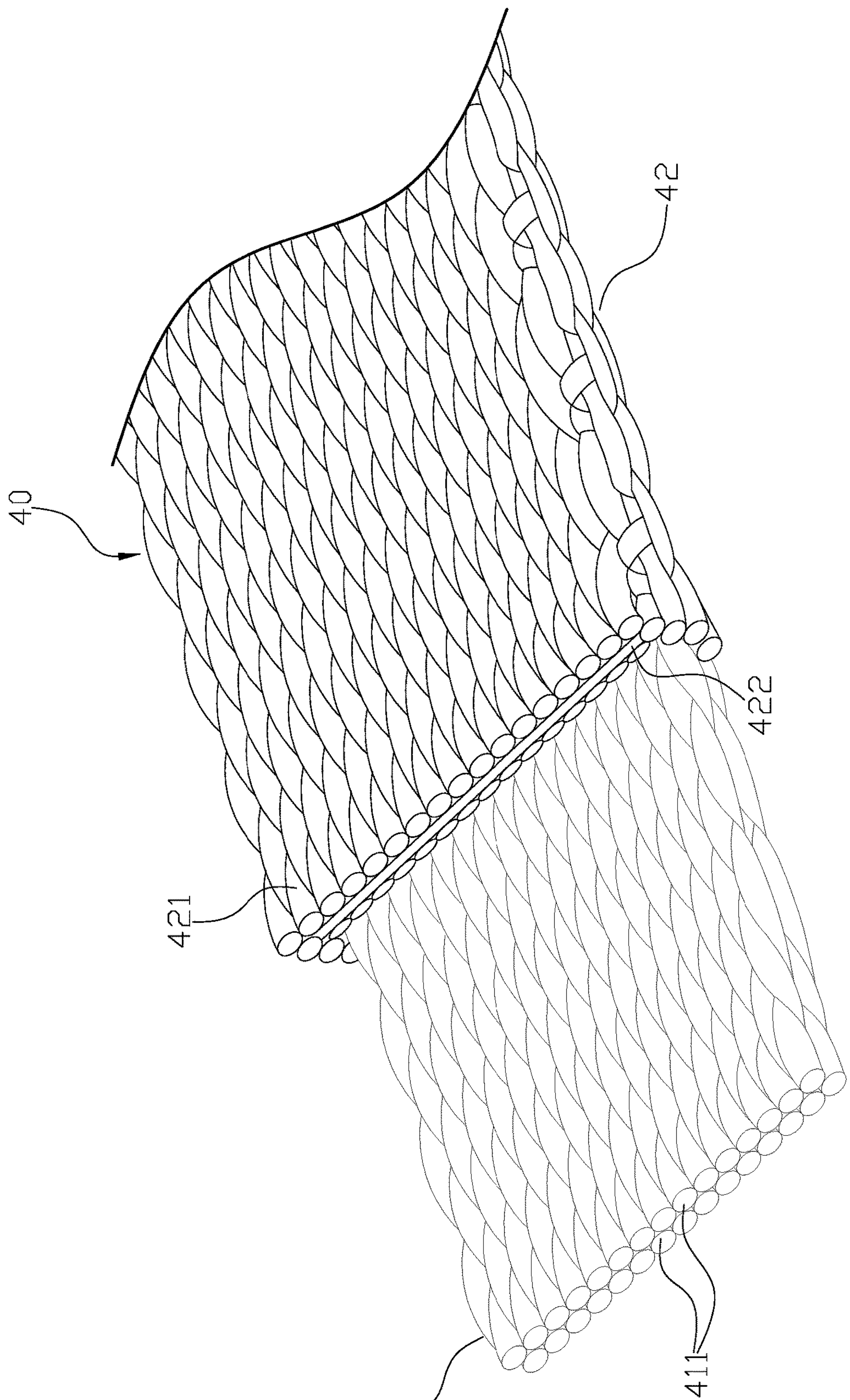


FIG. 8
PRIOR ART

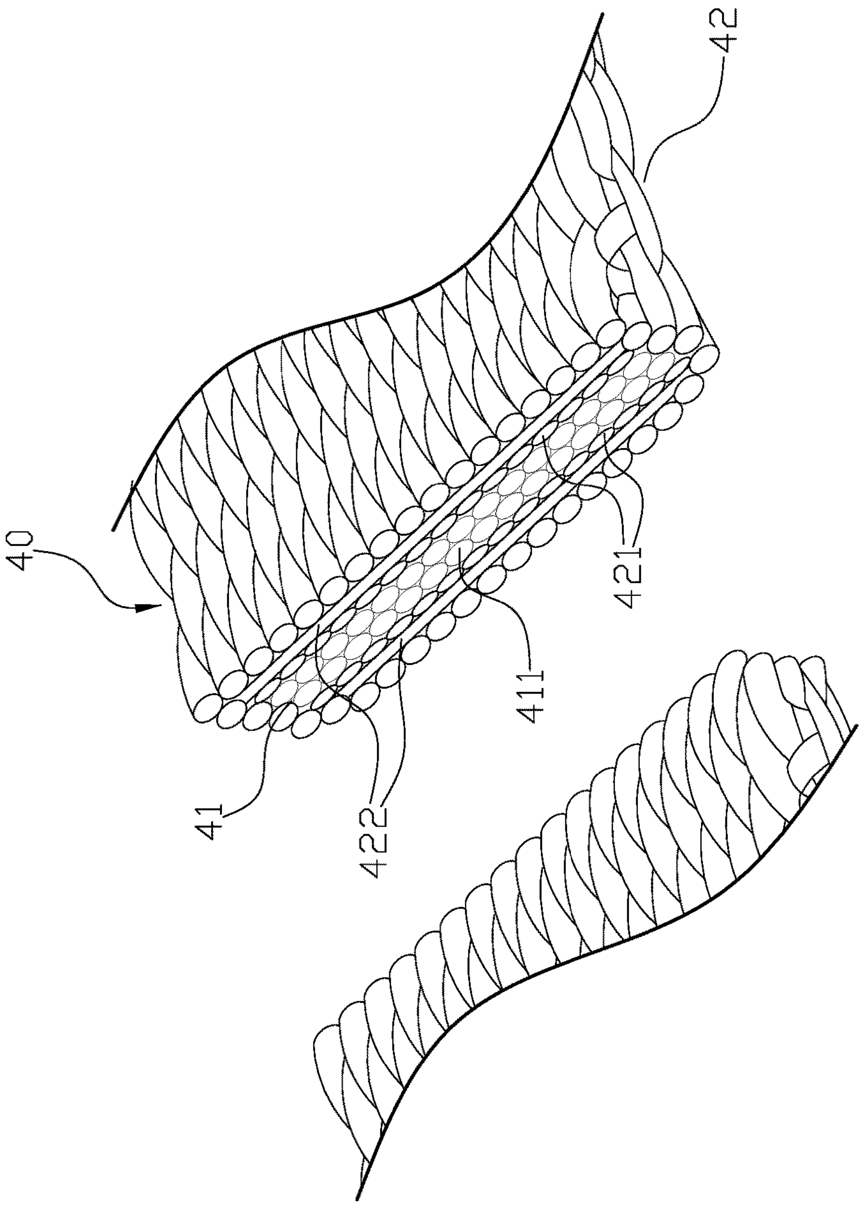


FIG. 9
PRIOR ART

SAFETY STRAP

BACKGROUND OF INVENTION

Field of Invention

The present invention relates to a safety strap.

Description of the Related Art

When working at high locations, personnel are required to wear anti-fall clothing or anti-fall belts on their bodies, so that they can be connected with anti-fall devices through the safety strap to protect personnel from working or moving at heights.

The conventional safety strap, as shown in FIG. **8**, the safety strap **40** includes the outer layer **42**, and the inner layer **41** wrapped in the outer layer **42**. The inner layer **41** is made of many warp strands **411**, and the outer layer **42** is made of many warp strands **421** and many weft strands **422**. The inner layer **41** is directly covered by the outer layer **42** to form a the safety strap structure.

However, the above conventional structure still has the following problem: the outer layer **42** only wraps the inner layer **41** inside, and not producing a reinforce effect on the inner layer **41**. Therefore, when the outer layer **42** is damaged due to the friction of use, such as being worn out by the sharp edge of a high-rise building, it can cause the warp strands **411** of the inner layer **41** to spread out from the damaged opening, allowing the safety strap **40** easy to break due to the loosen warp strands **411**, as shown in FIG. **9**.

Therefore, it is desirable to provide a safety strap to mitigate and/or obviate the aforementioned problems.

SUMMARY OF THE INVENTION

An objective of present invention is to provide a safety strap which is capable of improving the above-mention problems.

In order to achieve the above mentioned objective, a safety strap has an outer layer, an inner layer disposed in the outer layer and a plurality of interlace strands. The inner layer has a plurality of warp strand bundles and a plurality of first weft strands woven with each other. Each of the warp strand bundles has more than one first warp strand. The outer layer is woven with a plurality of second warp strands and a plurality of second weft strands. The interlace strands continuously intertwine with the second weft strands and the first weft strands.

Other objects, advantages, and novel features of invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a three-dimensional view of the connection between the present invention and a fall arrester.

FIG. **2** is the winding drawing of an embodiment of the present invention.

FIG. **3** is an enlarged view of the winding of the present invention.

FIG. **4** is a cross-sectional view of the present invention.

FIG. **5** is the state drawing showing the present invention accidentally dropping and rubbing against the building when it is in use.

FIG. **6** is a schematic drawing showing the friction occurring in the present invention.

FIG. **7** is a state drawing of the present invention that does not break after being rubbed.

FIG. **8** is a structural diagram of a prior art.

FIG. **9** is a diagram showing the fractured state after conventional friction.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

First, please refer to FIG. **1**-FIG. **4**. A safety strap **10** comprises an outer layer **12** an inner layer **11** disposed in the outer layer **12** and a plurality of interlace strands **20**. The inner layer **11** has a plurality of warp strand bundles **111** and a plurality of first weft strands **112** woven with each other. Each of the warp strand bundle **111** has more than one first warp strand **1111**. The outer layer **12** is woven with a plurality of second warp strands **121** and a plurality of second weft strands **122**. The interlace strands **20** continuously intertwine with the second weft strands **122** and the first weft strands **112**. With the interlace strands **20**, the inner layer **11** and the outer layer **12** are combined together and a reinforcement effect is provided to the inner layer **11**.

Furthermore, the warp strand bundle **111** are woven with three first warp strands **1111**.

Moreover, the warp strand bundles **111** are woven with three first warp strands **1111**, two of the three first warp strands **1111** respectively pass several first weft strands **112** from above and from below, and another first warp strand **1111** continuously passes around the first weft strand **112**.

Furthermore, the first warp strand **1111**, the second warp strand **121**, the first weft strand **112** and the second weft strand **122** are multifilament polymer strands, which are composed of UV-resistant liquid crystal aromatic polyester fibers, also known as LCP liquid crystal polymer aromatic polyester fiber.

In addition, the outer layer **12** is completely wrapped around the inner layer **11**.

For actual operation, the safety strap **10** is equipped with the anti-fall clothing or the anti-fall belt on a user's body, so that the anti-fall clothing or the anti-fall belt can be connected with the anti-fall device **30** via the safety strap **10**, as shown in FIG. **1**, to ensure the safety of personnel working at heights or activities. If the user accidentally falls during the operation or the second warp strand **121** of the outer layer **12** is worn out and damaged due to friction from the wall, the support or the iron sheet of the building, as shown in FIG. **5** and FIG. **6**, the warp strand bundle **111** can still be tightened continuously through the interlace strands **20**, and the first warp strands **1111** of the warp strand bundle **111** are not spread out at the damaged opening, therefore, the safety strap **10** can be kept in a tightened and not loose state because the warp strand bundle **111** is still in a tightened and not loose state and will not break, as shown in FIG. **7**. Also, the safety strap **10** has the tightening and strengthening effect of the inner layer **11**, so its structural strength can be significantly improved.

Although the present invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of invention as hereinafter claimed.

What is claimed is:

1. A safety strap comprising:

a top layer,

a bottom laver,

a middle layer disposed between the top layer and the bottom layer, and

a plurality of interlace strands;

wherein:

the middle layer comprises a plurality of warp strand bundles and a plurality of first weft strands woven with each other, each of the warp strand bundles comprising more than one first warp strand, each first warp strand consecutively float woven with groups of three first weft strands;

the top layer and the bottom layer are respectively woven with a plurality of second warp strands and a plurality of second weft strands; and

the interlace strands continuously intertwine with the second weft strands of the top layer, the second weft strands of the bottom layer, and the first weft strands;

wherein each interlace strand immediately and consecutively sandwiches each of the second weft strands of the top layer, each of the first weft strands, and each of the second weft strands of the bottom layer.

2. The safety strap as claimed in claim **1**, wherein the warp strand bundles are woven with three first warp strands.

3. The safety strap as claimed in claim **1**, wherein each of the warp strand bundles comprises three first warp strands, two of the three first warp strands respectively pass several first weft strands from above and from below, and another first warp strand sequentially passes above and below the first weft strand.

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